Calculations for the preferred design (Design A): solar powered harbour buoy

1. Assumptions

- Sizes and masses come from the CAD model (models/esp2024.py): all 43 parts are modelled with their own material density; the buoy has 77 parts and the sinker 5.
- Sea water 1025 kg/m³. The harbour lane is 5 m deep at high water; the buoy carries the weight of the chain hanging from it, about one depth of 10 mm chain (2.2 kg/m in air, 9.6 kg in water). Chain length 3 × depth.
- Wind for the heel check 17 m/s (gale force 8); tidal current 1.5 m/s; drag coefficient 1 for the float, panels and topmark.
- Sun: 1 kWh/m² a day in December and 3 in June on an upright panel facing the sun at 50° north; panels lose 25 % to salt, dirt, heat and the charger.
- Labour £17 an hour (marine fabrication). Material and bought in prices are supplier estimates for a batch of 100; they are listed in section 9.
- Marine growth, ice and a boat tying up to the buoy are not in the floating calculation; the reserve buoyancy covers them.

2. Mass and how the buoy floats

Quantity	Working	Result
Mass of the buoy (red), CAD model	sum of volume × density for all 77 parts	**173.4 kg**
Weight carried: buoy + hanging chain	173.4 + 9.6	183.0 kg
Sea water it must push aside	183.0 / 1025	0.1785 m³
Waterline, moored (from the CAD shape below the water)	found by trial: the volume below the line = the volume needed	**+1 mm** on the design waterline
Height of the top above the sea, red (can) buoy	1676 − (1) (brief 1400 to 1800)	**1675 mm**
Height of the top above the sea, green (cone) buoy	1696 − (-1)	**1697 mm**
If the chain breaks (no chain weight)	lighter, the buoy rises 48 mm: top 1723 mm above the sea	**still in range**
Base diameter at the waterline	float Ø500; over the fender Ø524 (brief 500 ± 25)	**500 mm**
Draught (waterline to the bottom of the mooring eye)	needs at least 2.3 m of water at low tide	1761 mm
Reserve buoyancy (pushed fully under)	0.2286 m³ × 1025 − 183.0	51 kg (28 %)

Sub-assembly	Mass
Float and deck	42.6 kg
Equipment canister: battery, control and tracking	21.0 kg
Tail tube, ballast and anode	85.8 kg
Tripod mast and solar panels	17.4 kg
Lantern and topmark	5.2 kg
Mooring, alert load pin and chain	1.5 kg

The ballast (86 kg with the tail) is the heaviest group on purpose: low down, it keeps the buoy upright. The sinker on the harbour bed weighs 301 kg in air (172 kg in water) and holds the buoy with the chain lying on the bed.

3. Stability: does it stay upright?

Quantity	Working	Result
Centre of buoyancy B (height)	centroid of the CAD shape below the waterline	-545 mm
Centre of gravity G, with the chain	from every part's mass and centre; chain weight at the load pin	-757 mm
BM = I / V	$I = \pi \times 500^4 / 64 = 3.06796\ m^4$; / 0.1785 m³	17 mm
Metacentric height GM = B + BM − G	-545 + 17 − (-757); positive, so it is stable	**229 mm**
GM if the chain breaks	-573 + 18 − (-704)	**149 mm**
Wind force on the buoy, 17 m/s	½ × 1.23 × 17² × 1 × 0.555 m² (table below)	99 N
Heeling moment	99 N × 1.33 m (wind centre to water drag centre)	131 N m
Righting moment at 90°: W × GM	183.0 × 9.81 × 0.229	412 N m
Steady heel in the gale	sin θ = 131 / 412	**18.5°**

Area above the water (side view)	Area	Centre height
float and fender	0.131 m²	125 mm
solar panels (one square on, two at 60 degrees)	0.240 m²	600 mm
tripod legs and struts	0.113 m²	838 mm
lantern	0.019 m²	1263 mm
topmark (can)	0.052 m²	1546 mm

4. Renewable energy: what the alert and tracking system needs (12 V)

Load (all 12 V DC)	Power on	Hours a day	Energy a day	How it is used
LED lantern, flashing Fl(2) 6 s	1.00 W	2.67 h	2.67 Wh	on 1 s in every 6 s (2 flashes of 0.5 s), 16 h of darkness
Lantern light sensor and driver, standby	0.02 W	24.00 h	0.48 Wh	all day
GNSS position fix and LTE-M report	0.12 W	24.00 h	2.88 Wh	a fix and report every hour: 1.4 W for about 5 minutes, averaged
Alert system: load pin and logic	0.24 W	2.40 h	0.58 Wh	load pin read for 1 s in every 10 s (20 mA at 12 V)
Control unit and MPPT charger, standby	0.05 W	24.00 h	1.20 Wh	all day
Total used a day (winter)			**7.8 Wh**	
Needed from the panels, with 20 % battery and wiring loss			**9.4 Wh**	**7.8 × 1.2**

5. Renewable energy: what the solar panels make

Quantity	Working	Result
Panels	3 × 20 W monocrystalline, upright, 120° apart round the mast	60 W
Panels working, sun square on to one panel	1 + 2 × 0.3 (the two facing away get diffuse light)	1.60 panels
Panels working, sun between two panels (the worst way round)	2 × cos 60° + 0.3	1.30 panels
Energy made a day in December (worst month)	20 W × 1 h × 0.75 × 1.30	**19.5 Wh**
Made against needed, December	19.5 / 9.4 (must be more than 1)	**2.1 ×**
Energy made a day in June	20 × 3 × 0.75 × 1.30: the charger stops when the battery is full	58.5 Wh
Battery	LiFePO4 12.8 V × 40 Ah = 512 Wh, use 80 %	410 Wh
Days with no sun at all	410 / 9.4	**44 days**
Drifting with the alert on (position every minute)	1 W × 24 h × 1.2 = 28.8 Wh a day; 410 / 28.8	14 days to find it

The supply is 12 V nominal (12.8 V LiFePO4, 10 to 14.4 V), which suits the lantern, tracker and load pin amplifier. Upright panels collect more in winter, when the sun is low, than flat ones, shed rain, spray and bird droppings, and cannot be stood on. The charger switches the lantern on at dusk from the panel voltage, so there is no separate light sensor to fail.

6. Mooring loads and the mooring eye

Quantity	Working	Result
Current drag on the parts under water	½ × 1025 × 1.5² × 1 × 0.549 m²	633 N
Wind drag (17 m/s)	section 3	99 N
Chain weight on the eye	9.6 kg × 9.81	94 N
Steady pull on the mooring eye, gale and tide together	√((633 + 99)² + 94²)	**738 N**
Reserve buoyancy (extra down force before the deck goes under)	51 kg × 9.81	504 N
Worst case: the tide lifts all the chain off the bed	3 × 5 m × 2.2 kg/m × (1 − 1025/7850) × 9.81, less the 94 N already carried	188 N, less than 504 N
Design load for the mooring eye	breaking load of the chain: the buoy must never be the weak link	**40 kN**
Plate above the hole (gross section)	40000 / (90 × 20)	22.2 MPa (FEA 20.8)
Net section beside the hole	40000 / ((90 − 30) × 20)	33.3 MPa (FEA 15.6)
Pin bearing on the hole	40000 / (28 × 20)	71.4 MPa
Tear out below the hole (two shear planes)	40000 / (2 × (45 − 15) × 20)	33.3 MPa (shear)
Fillet weld round the top of the eye, 6 mm leg	40000 / (2 × (90 + 20) × 0.7 × 6)	43.3 MPa
FEA peak (von Mises), at the edge of the hole	gmsh + scikit-fem, 10 node tetrahedra; 316L yield 170 MPa	**61 MPa**, safety factor **2.8**

The steady pull in a gale with the tide (738 N) is less than 2 % of the design load. It is mostly sideways: with the chain 3 times the depth it leaves the buoy at a shallow angle, and even if the strongest tide lifts all the chain off the bed the buoy carries only 188 N more, against 504 N of reserve buoyancy: it dips about 95 mm but stays afloat. Even at the full breaking load of the chain the mooring eye stays below yield: the chain fails first, the load pin sees the pull fall to zero and the alert is sent. Above the hole the FEA (20.8 MPa) agrees with the hand value (22.2 MPa). Beside the hole the FEA is about half the hand value: like Inventor, it spreads the force round the whole hole, so half of it goes straight into the plate above. The real pin bears on the bottom of the hole, so the hand value is the one to design to, and it is well below yield. The FEA peak at the edge of the hole is the stress concentration (about 2 to 3 times the net stress).

7. Cost to make one buoy

Cost element (one buoy with its sinker and chain)	Working	Result
Raw materials	mass × buy to fly × £/kg, + 5 % scrap (section 9)	£575
Bought in parts	battery, control unit, panels, lantern, load pin, fasteners, chain 3 × 5 m (£60)	£832
Labour	15.4 h making parts + 4 h assembly and test = 19.9 h × £17	£339
Factory overhead	25 % of labour	£85
Tooling share	£10,000 / 100	£100
Prototypes and sea trial share	£6,000 / 100	£60
Cost to make one buoy		**£1,990**
Budget (brief)	£200,000 / 100 buoys	£2,000
Margin	**£2,000 − £1,990**	**£10**
Whole order	£1,990 × 100 (budget £200,000)	£199,034

The biggest costs are the control and tracking unit, the battery, the load pin and the lantern: the electronics are about a third of the cost of the buoy. Stainless steel is used only where strength is needed; the float, topmark and ballast are cheap, corrosion free polyethylene and concrete.

8. Time: design, prototype and 100 buoys in 70 weeks

Stage	Start week	End week	Weeks
Design and drawings (this task), client approval	0	6	6
Rotomoulding tools and jigs made	4	10	6
Two prototypes built and tested in the tank	6	12	6
Sea trial of the prototypes in a client harbour	12	17	5
Changes from the trial, production release	17	18	1
Production: 100 buoys at 2 a week	18	68	50
Finished		**68**	**2 weeks spare**

Production: 100 buoys at 2 a week = 50 weeks (brief). Labour 19.9 h a buoy × 2 = 40 h a week: about 1.3 people at 80 % efficiency. Floats are rotomoulded two a day, so moulding never holds up the line.

9. Materials, corrosion and prices

Material	£/kg	Buy to fly	Why it was chosen
316L stainless steel plate, tube and bar	£4.20	1.25	Does not rust in sea water; protected by the anode
MDPE rotomoulding powder, UV stabilised	£3.50	1.10	Colour all through; recyclable
Closed cell PU foam, two part	£6.00	1.10	Unsinkable if the hull is holed
EVA foam fender with rubber skin	£14.00	1.10	Soft: takes minor impacts
HDPE sheet	£4.00	1.30	Does not corrode
Heavy concrete (barytes) in a PE skin	£0.80	1.05	Cheap, dense, cannot rust
Reinforced concrete C32/40	£0.12	1.05	Sinker: no maintenance

- No steel that can rust: the stainless is 316L (with molybdenum, for sea water), the fasteners A4 (316), the float, topmark and fender polyethylene and rubber, the ballast and sinker concrete.
- The zinc anode (22) is wired to the stainless tail and the galvanised chain: zinc is more reactive, so it corrodes instead of them. It is changed at the yearly service.
- Nylon washers and sleeves keep the galvanised chain off the stainless eye where they are not joined through the anode, and all stainless parts are passivated after welding.

10. Mass and cost of every part (one red buoy with its sinker)

Item	Part	Qty	Mass kg	Material	Material	Hours	Bought in
1	float hull	1	13.57	MDPE rotomoulding powder, UV stabilised	£55	0.75	–
2	float foam fill	1	6.04	Closed cell PU foam, two part	£42	0.30	–
3	fender ring	1	0.87	EVA foam fender with rubber skin	£14	0.20	–
4	deck plate	1	5.18	316L stainless steel plate, tube and bar	£29	0.60	–
5	keel plate	1	10.99	316L stainless steel plate, tube and bar	£61	0.50	–
6	tie rod	3	5.14	Bought in	–	0.15	£12
7	tie rod nut M16	3	0.09	Bought in	–	0.00	£2
8	lifting eye nut M16	3	0.70	Bought in	–	0.15	£27
9	equipment canister	1	9.10	316L stainless steel plate, tube and bar	£50	1.20	–
10	equipment mounting plate	1	0.43	HDPE sheet	£2	0.20	–
11	battery 12.8 V 40 Ah	1	3.69	Bought in	–	0.10	£140
12	battery strap	1	0.25	316L stainless steel plate, tube and bar	£1	0.20	–
13	control and tracking unit	1	1.71	Bought in	–	0.50	£150
14	hatch O-ring	1	0.02	Bought in	–	0.00	£6
15	hatch lid	1	5.58	316L stainless steel plate, tube and bar	£31	0.60	–
16	tamper proof bolt M10	8	0.16	Bought in	–	0.00	£12
17	cable gland M16	2	0.02	Bought in	–	0.20	£4
18	tail tube	1	14.41	316L stainless steel plate, tube and bar	£79	1.50	–
19	hex bolt M12 x 20	6	0.22	Bought in	–	0.00	£4
20	ballast ring	1	64.85	Heavy concrete (barytes) in a PE skin	£57	0.50	–
21	ballast collar	1	1.96	316L stainless steel plate, tube and bar	£11	0.30	–
22	zinc anode	1	3.17	Bought in	–	0.10	£28
23	mooring eye	1	1.17	316L stainless steel plate, tube and bar	£6	0.40	–
24	tripod leg	3	5.50	316L stainless steel plate, tube and bar	£30	1.80	–
25	hex bolt M12 x 16	6	0.19	Bought in	–	0.00	£4
26	mast top plate	1	2.41	316L stainless steel plate, tube and bar	£13	0.30	–
27	lower panel bracket	3	2.10	316L stainless steel plate, tube and bar	£12	0.75	–
28	upper panel bracket	3	2.31	316L stainless steel plate, tube and bar	£13	0.75	–
29	solar panel 20 W	3	4.93	Bought in	–	0.60	£84
30	lantern base	1	0.41	Bought in	–	0.30	£110
31	lantern lens	1	0.27	Bought in	–	0.00	£20
32	lantern cap	1	0.15	Bought in	–	0.00	£8
33	topmark strut	3	1.00	316L stainless steel plate, tube and bar	£6	0.60	–
34	topmark plate	1	2.35	316L stainless steel plate, tube and bar	£13	0.30	–
35	GNSS and mobile antenna	1	0.10	Bought in	–	0.10	£30
36	topmark, can (red)	1	0.89	MDPE rotomoulding powder, UV stabilised	£4	0.50	–
38	bow shackle	1	0.29	Bought in	–	0.00	£14
39	load pin (alert sensor)	1	0.36	Bought in	–	0.20	£95
40	mooring swivel	1	0.53	Bought in	–	0.00	£22
41	chain link 10 mm	7	0.57	Bought in	–	0.00	–
42	concrete sinker	1	299.60	Reinforced concrete C32/40	£40	0.50	–
43	sinker eye bar	1	1.19	316L stainless steel plate, tube and bar	£7	0.30	–

The green buoy is the same, with the cone topmark (37) in place of the can (36): mass 173.2 kg, the same cost. The 7 chain links are part of the chain, costed in section 7.